

ABSTRACT

Nowadays Women Safety is the biggest concern in many parts of the world. There is still a fear in alone areas for women as well as men. So here we propose a security patrolling robot using ESP32CAM. The system uses camera mounted on robotic vehicle for securing any premises. It monitors each area to detect any problem using camera with remote location. It has the ability to monitor live streaming video in the premises. It provides continuous monitoring together with live broadcasting of the site off crime. This robot is controlled by with mobile or laptop with IP address of robot. The robot control will be with (left, right, forward, backward and stop). We can alert the robot surroundings with buzzer

LIST OF CONTENTS

S. No	CONTENTS	PAGE NO.
	ABSTRACT	I
	LIST OF CONTENTS	II
	LIST OF FIGURES	IV
	LIST OF TABLES	V
	ABBREVIATIONS	VI
1	INTRODUCTION	1
	1.1 objective	2
2	LITERATURE SURVEY	3
	2.1 A Mobile based Women Safety Application	3
	2.2 Abhaya An Android App for The Safety of Women	3
	2.3 Violence Against Homeless Women	4
	2.4 Smart Wearable Device for Women Safety Using Iot	4
3	EXISTING SYSTEM	5
	3.1 Drawbacks	5
4	PROPOSED SYSTEM	6
	4.1 Benefits	6
	4.2 Proposed Block Diagram	7
	4.3. HARDWARE TOOLS	7
	4.3.1 ESP32 Module	8
	4.3.2 L293D Driver	9
	4.3.3 Motors	11
	4.3.4 Camera	13
	4.3.5 Buzzer	14
	4.3.6 Regulated Power Supply	16
5	SOFTWARE TOOLS	25
	5.1 Arduino Ide	25

	5.2 Code	31
6	RESULTS and DISCUSSION	37
7	CONCLUSION	41
8	FURURE SCOPE	42
9	REFERENCES	43

LIST OF FIGURES

S.NO	Name of The Figure	Page. No
1	Block Diagram of Women Safety Night Patrolling Robot	7
2	Block Diagram of ESP32	8
3	Block Diagram of L293D	10
4	Dc Motor	12
5	Construction of DC Motor	12
6	Working of DC Motor	13
7	Camera	14
8	Buzzer	15
9	Voltage Buzzer Controls	16
10	Block Diagram of Regulated Power Supply	16
11	Types of Transformers	17
12	Half Wave Rectifier	18
13	Full Wave Rectifier	19
14	Bridge Rectifier	20
15	Bridge Rectifier with Centre Tapped	20
16	Switching Regulator	23
17	Circuit Diagram of Power Supply	23
18	Women Safety Night Patrolling Robot	37
19	Image of Robot When night Light Is On	38
20	Control Panel of Robot and Live Streaming Video	39
21	Controlling The Robot with Mobile Phone	40

LIST OF TABLES

S. No	Name of the Tables	Page. No
1	Specifications of IC7805	24

LIST OF ABBREVIATIONS

RF	Radio Frequency
DC	Direct Current
IOT	Internet of Things
LED	Light Emitting Diode
IDE	Integrated Development Environment
RPS	Regulated power supply
GPIO	General Purpose I/O Pins
PSRAM	Pseudo Static Random Access Memory